

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Capacitor equation rearranged $d = \frac{\epsilon_0 A}{C}$ (1) Substitution of values (1) Correct answer 1.43×10^{-4} m (0.143 mm) (1) [Accept 1.4×10^{-4} m, not 1×10^{-4}]	1	1 1		3	3	
		(ii)	Adding capacitors in parallel (4 nF) (1) Understanding of 6 nF in series with 4 nF i.e. correct application of physics (1) [even failure to invert to get answer, so $1/4 + 1/6 \rightarrow 2$ marks] Correct answer 2.4 n[F] or 12/5 n[F] (1) c.a.o.	1	1 1		3	2	
		(iii)	Any valid CV e.g. 2×9 , 6×6 , 2.4×15 (1) [Accept 2.4×15 or $\frac{1}{2} \times 2.4 \times 15$; not 1.2×15 or 2.4×7.5] $\frac{1}{2}$ -way explanation: charge split / divided between 2 nFs / 9 V across 2 nF / 6 V across 6 nF (1) Completed explanation: all charge on capacitors in series / justify 9 V or 6V [e.g. $V \propto 1/C$]	1	1 1		3	2	
		(iv)	<u>Correct</u> substitution (into valid equation) e.g. 0.5 QV etc. (1) Correct answer (270 nJ or 2.7×10^{-7} J) (1)	1	1		2	2	
	(b)	(i)	Capacitor in series with resistor and cell/battery/psu (1) Properly placed ammeter and voltmeter (1)	2			2		2

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		(ii)	Emf = 12 V stated [accept $V_{\max}$] (1) Initial current = 2.7 mA (1) $R = \frac{V}{I} = 4\,444\,\Omega$ (1) [4.4 kΩ] [N.B. With incompatible V and I, 2nd and 3rd marks not accessible] $RC \sim 8$ s from 37% $I_{\max}$ value or 63% $V_{\max}$ value or intercept of gradient at the origin with $I = 0$ (or $V = 12$ V) OR taking logs and rearranging OR obtaining Q from total area under current graph around 2.0×10^{-2} C (1) $C = \frac{8.0}{4400} = 1.8$ [± 0.3] mF (1) [No ecf] Unit penalty (–1): all 3 units [V, Ω, F] needed Alternative Emf = 12 V (1) Find $\frac{dV}{dt}$ and I at given t, e.g. at 10 s, 0.411 V s^{–1}, 7.5 mA (1) Hence calculate C from $I = C \frac{dV}{dt} \rightarrow 1.8$ mF ecf (1) $RC \sim 8$ s from 37% $I_{\max}$ value or 63% $V_{\max}$ value or intercept of gradient at $t = 0$ with $I = 0$ (or $V = 12$ V) (1) R from C and $RC \rightarrow 4.4$ kΩ (1) Unit penalty (–1): all 3 units [V, Ω, F] needed		5		5	5	5
			Question 1 total	6	12	0	18	14	7